PV102 Day 7: Installation Planning Mechanical Layout and JHA

Students plan an installation before touching hardware: layout, material staging, wire routes, access, roof or ground hazards, tool control, quality checks, and job hazard analysis.

PV102 | CTS Day 23 | Instructor Run Sheet
Plan the install before lifting parts. | Wednesday, August 12, 2026

Core rule

Plan the install before lifting parts.

Learning targets

- Convert a design into a sequenced work plan.
- Identify mechanical, electrical, fall, weather, access, and tool hazards.
- Build a material/tool list that matches the scope.
- Use checklists to prevent skipped steps and hidden coupling.

Readiness and materials

Before students arrive	Materials
Open slide deck and test display. Print student packets and one instructor key. Prepare three team stations. Review safety boundary before any equipment is handled.	Mock roof or tabletop layout Rail/bracket cards or hardware JHA template Tool cards Tape measure Markers Projector and slide deck

Five-hour run of show

Time	Block	Instructor move
10:00-10:15	Opening and continuity	Connect Day 6 to today's rule. Name what students already know and what new judgment is being added.
10:15-10:40	Morning quiz and review	Students work independently for 12-15 minutes. Review by asking for evidence, not just letters.
10:40-11:25	Concept lesson	Use slides 1-14. Pause after each technical rule and ask what decision it changes.
11:25-12:15	Guided practice	Model one example on the board. Students complete a quick pair-check before lunch.
12:15-12:45	Lunch/reset	Reset lab stations. Keep energized work out of scope unless explicitly authorized by instructor and equipment procedure.
12:45-1:00	Lab brief	Assign teams and roles. Review acceptance criteria and stop-work

		triggers.
1:00-2:15	Install Plan and Job Hazard Analysis	Teams create a one-day install plan for a training system and defend the safety controls before work begins.
2:15-2:40	Handoffs and peer verification	Each team gives a concise technical handoff. Peers check evidence, boundary, and unknowns.
2:40-3:00	Exit ticket and reset	Collect packets. Preview next step: PV102 Day 8 uses commissioning and troubleshooting to verify the system and complete a professional handoff.

Instructor talk track

Opening script: "PV102 is not about throwing parts at a problem until reality files a complaint. Today we use installation planning mechanical layout and jha to make the next design decision safer and more defensible. The rule is simple: Plan the install before lifting parts."

Teaching notes

Move	Look-for
Ask	What decision does this evidence change?
Listen for	Students separating facts, assumptions, and unknowns.
Correct gently	Component-first answers. Redirect to job, evidence, boundary, then component.
Require	A handoff that another person could act on without reading minds. Humanity keeps trying telepathy. It remains a poor project-management system.

Lab instructor procedure

- Choose a scenario and define the scope in one sentence.
- Create a sequenced work plan: setup, layout, mounting/staging, wiring route, checks, cleanup.
- Complete a JHA with hazards, controls, PPE, and stop-work triggers.
- Build a material/tool list and mark missing items.
- Add at least three quality hold points.
- Handoff the plan as if a substitute crew had to run it safely.

Answer snapshot and contingency plan

If equipment or printing is unavailable, run the lab with paper cards, chart paper, and oral handoffs. The learning target is decision quality, not hardware theater.

Safety and scope boundaries

No live electrical work is authorized by this lesson unless the instructor independently sets up a verified training system, follows manufacturer instructions, and establishes the required isolation, PPE, and supervision. Treat PV source circuits and batteries as energy sources, not props.

Day handoff

PV102 Day 8 uses commissioning and troubleshooting to verify the system and complete a professional handoff.

Instructional sources

- U.S. Department of Energy, Solar Energy: https://www.energy.gov/topics/solar-energy
- U.S. Department of Energy, Solar Photovoltaic Technology Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-technology-basics
- U.S. Department of Energy, Solar Photovoltaic System Design Basics: https://www.energy.gov/eere/solar/solar-photovoltaic-system-design-basics
- U.S. Department of Energy, Solar Integration: Solar Energy and Storage Basics: https://www.energy.gov/eere/solar/solar-integration-solar-energy-and-storage-basics
- U.S. Department of Energy, Solar Integration: Inverters and Grid Services Basics: https://www.energy.gov/eere/solar/solar-integration-inverters-and-grid-services-basics
- NREL PVWatts Calculator: https://pvwatts.nrel.gov/
- OSHA, Green Job Hazards - Solar Energy: Electrical: https://www.osha.gov/green-jobs/solar/electrical
- OSHA, Green Job Hazards - Solar Energy: Lockout/Tagout: https://www.osha.gov/green-jobs/solar/lockout-tagout
- OSHA, Green Job Hazards - Solar Energy: Personal Protective Equipment: https://www.osha.gov/green-jobs/solar/personal-protective-equipment
- Sean White, Solar Photovoltaic Basics, 2nd ed., Routledge, 2018, ISBN 9781138102866.

PV102 Day 7 Morning Quiz: Answer Key

10 questions | 15 minutes | Independent work

PV102 | CTS Day 23 | Morning Quiz - Answer Key
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Directions: Answer from your current knowledge. For written responses, use complete technical statements. If unsure, state the safe next action rather than guessing.

1. A JHA is used to:

- A. Identify hazards and controls before work
- B. Replace all training
- C. Make electricity harmless
- D. Skip PPE

Answer: A

Rationale/look-for: Job hazard analysis names hazards and controls before the task begins.

2. A work sequence should include:

- A. Setup, controls, task steps, checks, cleanup
- B. Only the fastest path
- C. Only tool names
- D. No safety boundary

Answer: A

Rationale/look-for: Sequencing prevents unsafe improvisation.

3. The best time to plan material staging is:

- A. Before work starts
- B. After parts are missing
- C. During an emergency
- D. Never

Answer: A

Rationale/look-for: Staging avoids wasted motion and unsafe clutter.

4. A wire route should consider:

- A. Protection, distance, access, and serviceability
- B. Wall color only
- C. The shortest random path
- D. Nobody knows

Answer: A

Rationale/look-for: Wire route choices affect safety, voltage drop, damage risk, and maintenance.

5. A quality hold point means:

- A. Stop and verify before covering or continuing
- B. Stop learning
- C. Hide errors
- D. Energize faster

Answer: A

Rationale/look-for: Hold points catch defects before they become buried or dangerous.

6. Name one mechanical hazard in PV installation planning.

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept falls, lifting, cuts, pinch points, dropped objects, roof damage, weather, ladder access.

7. Name one electrical hazard that remains even before final commissioning.

_____

_____

Answer: Instructor judgment

Rationale/look-for: Accept PV live in light, stored battery energy, wrong polarity, short circuit, energized adjacent circuits.

8. Why should tools and materials be staged instead of scattered?

_____

_____

Answer: Control

Rationale/look-for: Prevents trips, loss, damage, time waste, and skipped checks.

9. What information from Day 2 changes the install plan?

_____

_____

Answer: Site/access/route

Rationale/look-for: Accept access, roof condition, wire route, shade objects, structure, weather exposure, mounting area.

10. Write the Day 7 rule in your own words.

_____

_____

Answer: Plan before work

Rationale/look-for: Look for planning before lifting/installing.

PV102 Day 7 Whiteboard Outline

PV102 | CTS Day 23 | Whiteboard Outline
Plan the install before lifting parts. | Wednesday, August 12, 2026

Board 1: Job	Board 2: Evidence	Board 3: Handoff
Topic: Installation Planning Mechanical Layout and JHA Rule: Plan the install before lifting parts. Question: What job are we solving?	Measured / Observed / Unknown Constraint that matters most Safety boundary before work	Decision Evidence Risk Unknowns Next action

Scenario anchors

- Tabletop PV training kit build: no energized connections until inspection.
- Mock roof rack layout: rails, clamps, wire management, no live module connection.
- Bus conversion equipment board: battery/inverter/controller placement and service clearances.

Final board sentence

Plan the install before lifting parts.

PV102 Day 7 Student Packet

Installation Planning Mechanical Layout and JHA

PV102 | CTS Day 23 | Student Print Packet
Plan the install before lifting parts. | Wednesday, August 12, 2026

Today's rule

Plan the install before lifting parts.

Learning targets

- Convert a design into a sequenced work plan.
- Identify mechanical, electrical, fall, weather, access, and tool hazards.
- Build a material/tool list that matches the scope.
- Use checklists to prevent skipped steps and hidden coupling.

Morning quiz

1. A JHA is used to:

- A. Identify hazards and controls before work
- B. Replace all training
- C. Make electricity harmless
- D. Skip PPE

2. A work sequence should include:

- A. Setup, controls, task steps, checks, cleanup
- B. Only the fastest path
- C. Only tool names
- D. No safety boundary

3. The best time to plan material staging is:

- A. Before work starts
- B. After parts are missing
- C. During an emergency
- D. Never

4. A wire route should consider:

- A. Protection, distance, access, and serviceability
- B. Wall color only
- C. The shortest random path
- D. Nobody knows

5. A quality hold point means:

- A. Stop and verify before covering or continuing
- B. Stop learning
- C. Hide errors
- D. Energize faster

6. Name one mechanical hazard in PV installation planning.

_____

_____

7. Name one electrical hazard that remains even before final commissioning.

_____

_____

8. Why should tools and materials be staged instead of scattered?

_____

_____

9. What information from Day 2 changes the install plan?

_____

_____

10. Write the Day 7 rule in your own words.

_____

_____

Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

Install Plan and Job Hazard Analysis

Teams create a one-day install plan for a training system and defend the safety controls before work begins.

Scenario options

- Tabletop PV training kit build: no energized connections until inspection.
- Mock roof rack layout: rails, clamps, wire management, no live module connection.
- Bus conversion equipment board: battery/inverter/controller placement and service clearances.

Team procedure

- Choose a scenario and define the scope in one sentence.
- Create a sequenced work plan: setup, layout, mounting/staging, wiring route, checks, cleanup.
- Complete a JHA with hazards, controls, PPE, and stop-work triggers.
- Build a material/tool list and mark missing items.
- Add at least three quality hold points.
- Handoff the plan as if a substitute crew had to run it safely.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

What is one step your team added that prevents a mistake before it becomes expensive or unsafe?

_____

_____

PV102 Day 7 Afternoon Lab

Install Plan and Job Hazard Analysis

PV102 | CTS Day 23 | Afternoon Lab
Plan the install before lifting parts. | Wednesday, August 12, 2026

Mission

Teams create a one-day install plan for a training system and defend the safety controls before work begins.

Safety boundary

This lab is training work. Do not energize or connect equipment outside the instructor-approved procedure. When in doubt, stop, isolate, verify, and ask.

Scenario options

- Tabletop PV training kit build: no energized connections until inspection.
- Mock roof rack layout: rails, clamps, wire management, no live module connection.
- Bus conversion equipment board: battery/inverter/controller placement and service clearances.

Procedure

- Choose a scenario and define the scope in one sentence.
- Create a sequenced work plan: setup, layout, mounting/staging, wiring route, checks, cleanup.
- Complete a JHA with hazards, controls, PPE, and stop-work triggers.
- Build a material/tool list and mark missing items.
- Add at least three quality hold points.
- Handoff the plan as if a substitute crew had to run it safely.

Acceptance criteria

- Match the real job.
- Show the energy or work path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.
- Give a handoff another crew can use.

Team handoff form

Item	Response
Team members	
Scenario	
Primary decision	

Evidence	
Safety boundary	
Unknowns	
Next safest action	

PV102 Day 7 Classroom Post

Copy-ready announcement and assignment text

PV102 | CTS Day 23 | Google Classroom Post
Plan the install before lifting parts. | Wednesday, August 12, 2026

Announcement

Today in PV102 we are working on installation planning mechanical layout and jha. The core rule is: Plan the install before lifting parts.

Assignment title

PV102 Day 7 - Installation Planning Mechanical Layout and JHA

Student directions

- Complete the morning readiness quiz independently.
- Use the reference section during the afternoon lab.
- Work with your team to complete the scenario handoff.
- Complete the exit ticket before class ends.

Turn in

- Completed student packet
- Team lab handoff
- Exit ticket

Success criteria

Match the need. Show the path. Name the constraint. State the safety boundary. Record unknowns.